



Qatar University

College of Engineering

**Department of Computer Science and
Engineering**

CMPE 370 Computer Engineering Practicum L02
Course Project Report
Fall 2025

PROJECT TITLE

SMART STORE AUTOMATION SYSTEM

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Introduction

This project is centered around the implementation of a smart shop named NovaMart that utilizes several smart features that do not require manual intervention. It includes an automatic lighting system, door system, temperature-controlled fan, and a security alarm. Moreover, the shop's building is built using custom handiwork.

Its first feature is an automatic lighting system built using a light dependent resistor (LDR). The LDR is able to detect changes in the surrounding lighting and thus is used to light up a number of LEDs when it is dark. This is determined using a voltage divider of resistors that are compared with the LDR's value in a comparator operational amplifier.

Next, it includes an automatic door system. When motion is detected in front of the store's door using a passive infrared (PIR) sensor, a 555 timer triggers for 5 seconds to enable the gear motor attached to the door.

Another feature of NovaMart is a temperature-controlled fan. When the surrounding temperature is high, the fan will turn on to reduce it. Temperature is detected with the use of the LM35 sensor, and the threshold to trigger the fan is determined with a potentiometer in a comparator circuit.

The final smart feature is a security alarm system. When the switch is set, motion is detected using a PIR sensor that will trigger a buzzer for 5 seconds, which will continue for as long as motion is detected. This way, thieves can be caught after the store's closing time.

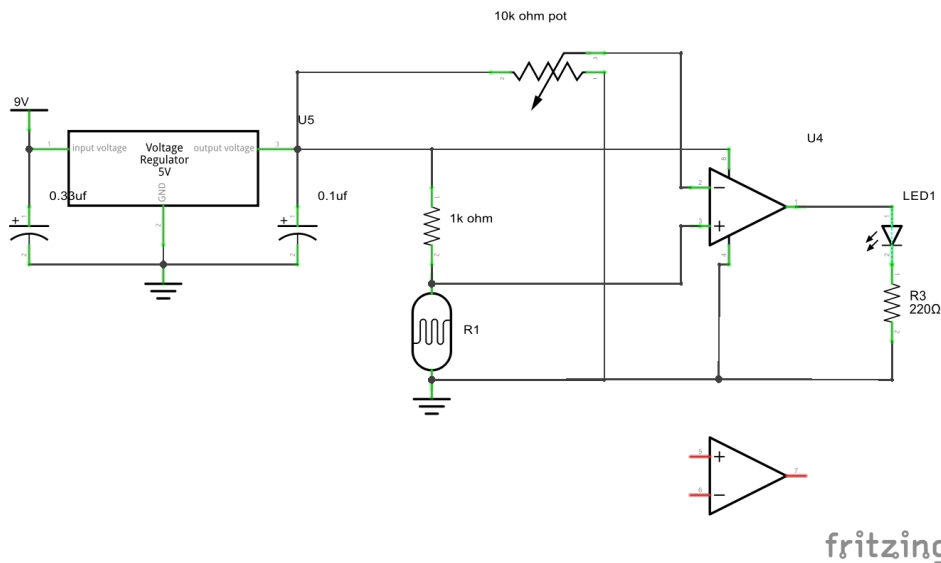
Power is supplied to all circuits used in the design through an LM7805 power regulator that receives power from a 9V battery.

With the combination of automated lighting, temperature control, motion detected entrance, and security alarm, NovaMart is able to implement numerous automated systems to serve both its customers and the store's needs.

Design

Automatic Lighting:

The automatic lighting circuit receives a constant 5V from the LM7805 regulator. The LM293D operational amplifier receives inputs from both the light dependent resistor as well as a potentiometer. When the surrounding is sufficiently dark, the LDR gives a high reading and so the LEDs turn on.

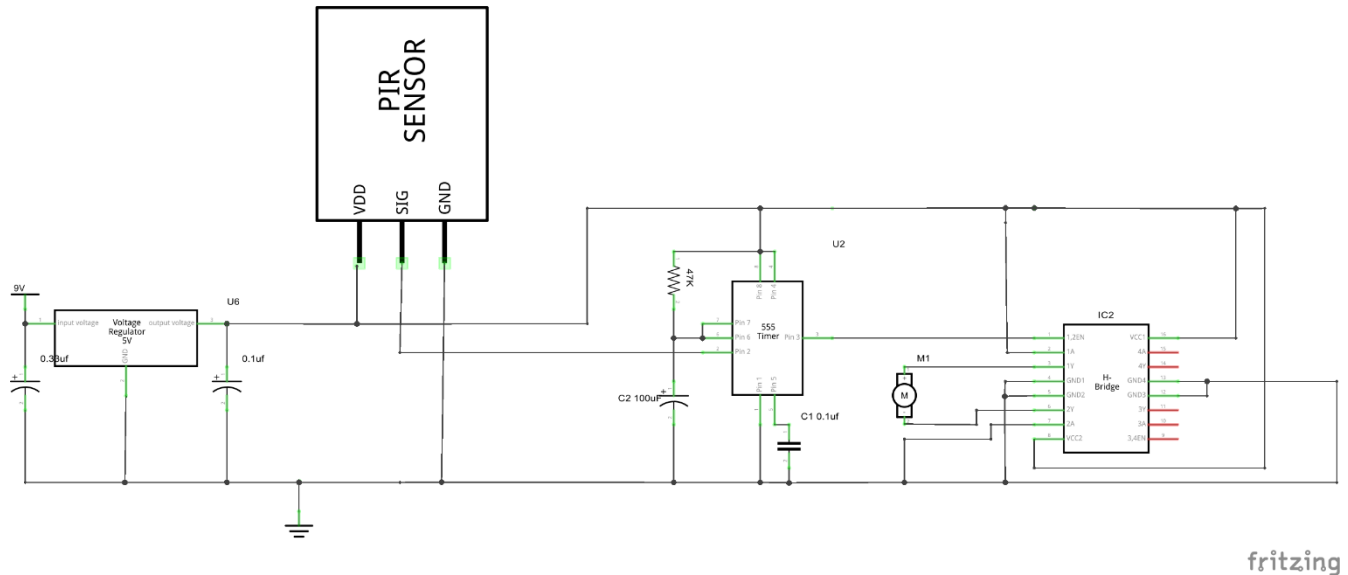


The comparator outputs 5V if $V_{LDR} > V_{pot}$, and 0 when $V_{LDR} < V_{pot}$.

V_{LDR} is around 1V or less when lights are on, and 4V or higher when there is a lack of light. V_{pot} is thus set to around 2.5V.

Automatic Door:

The PIR sensor gives a high reading when motion is detected, which will activate the 555 timer. The timer gives a high signal for 5 seconds, thanks to the resistor and capacitor values, turning on the motor for the door through the H-Bridge. Constant 5V is supplied through an LM7805 regulator.

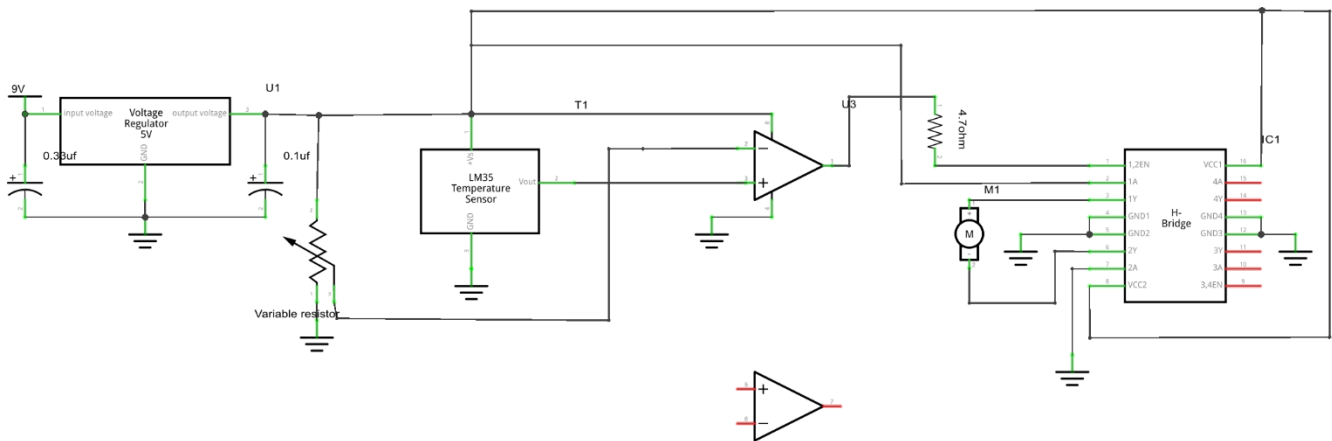


555 timer gives a high signal for a certain amount of time when PIR detects motion and thus outputs 0V. This causes trigger pin to fall below $1/3V_{cc}$ which turns on the output. As a result, the motor turns on for as long as the timer gives a high output to the H-Bridge.

Timer on-time: $T_{on} = 1.1RC = 1.1 \times 47k \times 100\mu = 5.17 s$

Temperature-Controlled Fan:

Voltage is supplied at a constant 5V from the LM7805 regulator. Temperature is detected through the LM35 sensor. When the surrounding is sufficiently hot, its voltage will be greater than that of the potentiometer, and thus the LM358 operational amplifier gives a high voltage to the H-Bridge, turning on the fan to cool the surroundings.



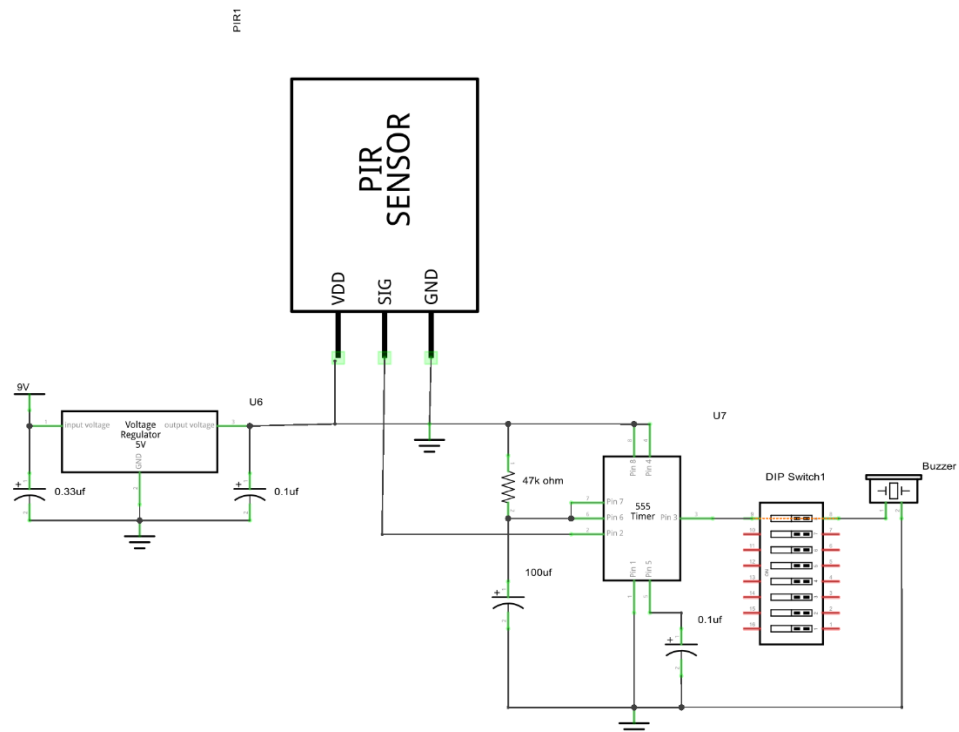
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LM35 voltage: $V_{LM35} = T_{\circ C} \times 10mV$, At $T=30$: $V_{LM35} = 30 \times 10mV = 300mV$

If $V_{LM35} > V_{pot}$ then the comparator outputs 5V. Otherwise, it outputs 0V. Thus, the potentiometer voltage should be set to the threshold of heat. Fan will turn on when comparator outputs 5V to reduce V_{LM35} to below V_{pot} .

Security Alarm:

When motion is detected, the PIR sensor gives a high reading to the 555 timer, which in turn turns on the buzzer for 5 seconds, thanks to the resistor and capacitor values. A DIP switch is used to prevent the buzzer from activating when it's turned off. All components receive necessary voltage from the LM7805 regulator, which supplies a constant 5V.



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555 timer goes high when trigger pin falls below $1/3V_{cc}$. As PIR sensor outputs 5V when idle and 0V when motion is detected, motion will activate the timer.

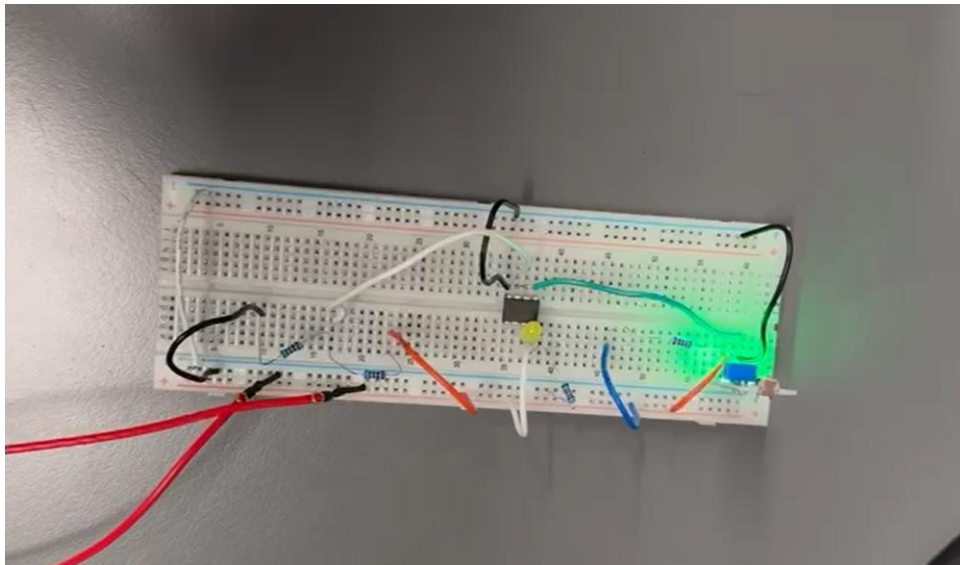
Timer on-time: $T_{on} = 1.1RC = 1.1 \times 47k \times 100\mu = 5.17 s$

Bill of Materials

Component	Description	Quantity
LM7085	5V Voltage Regulator	1
Battery	9V battery	1
LM35	Temperature sensor	1
LM358	Dual operational amplifier	2
DIP Switch	8 position DIP switch	1
Buzzer	Active buzzer	1
LED	Yellow LED	1
Resistor	1 k Ω	2
Resistor	47 k Ω	2
Resistor	220 Ω	1
Capacitor	0.1 μ F	3
Capacitor	0.33 μ F	1
Capacitor	100 μ F	2
Potentiometer	10k ohm	1
PIR	Motion detection sensor	2
LM293D	H-Bridge	2
Fan	DC Fan	1
Motor	Gear Motor	1
NE555N	555 timer	2
LDR	Light dependent resistor	1

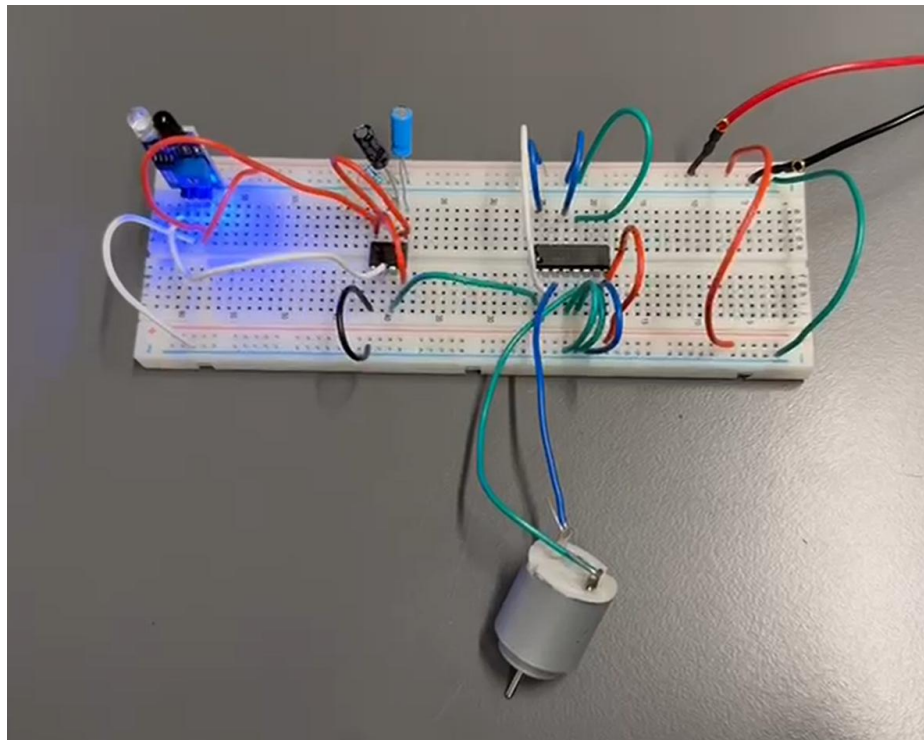
Breadboard Prototype & Testing

Automatic Lighting:



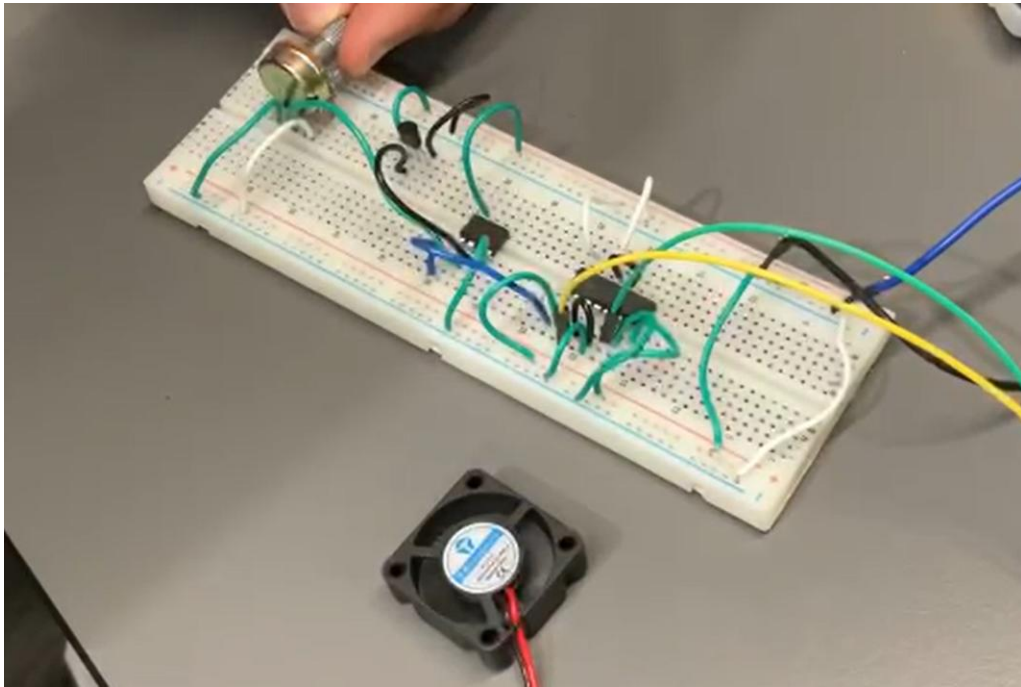
The LED turns on when the LDR is covered and remains off when uncovered.

Automatic Door:



The motor activates for around 5 seconds when motion is detected.

Temperature-Controlled Fan:



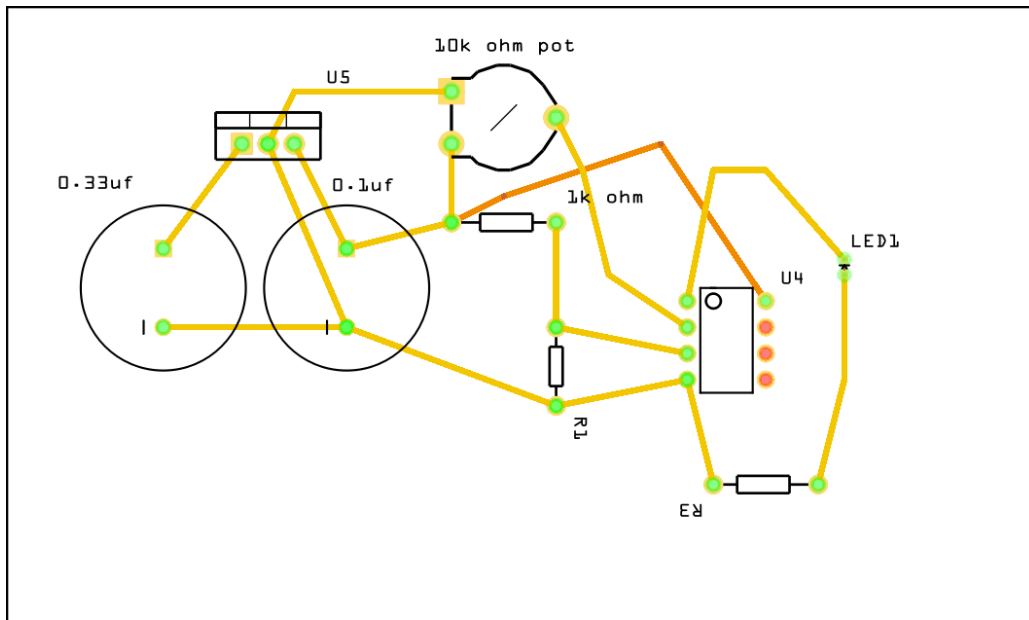
The fan turns on when the threshold of the temperature as controlled by the potentiometer is exceeded.

Security Alarm:

The buzzer sounds when motion is detected while switch is enabled. Otherwise, buzzer makes no sound even when motion is detected.

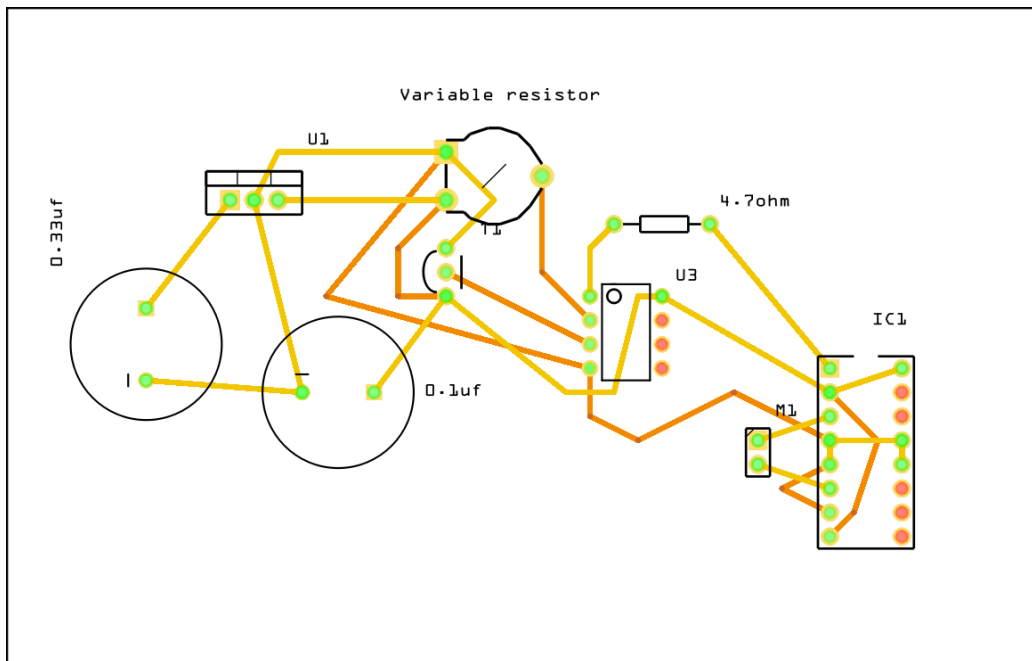
PCB Design

Automatic Lighting PCB:



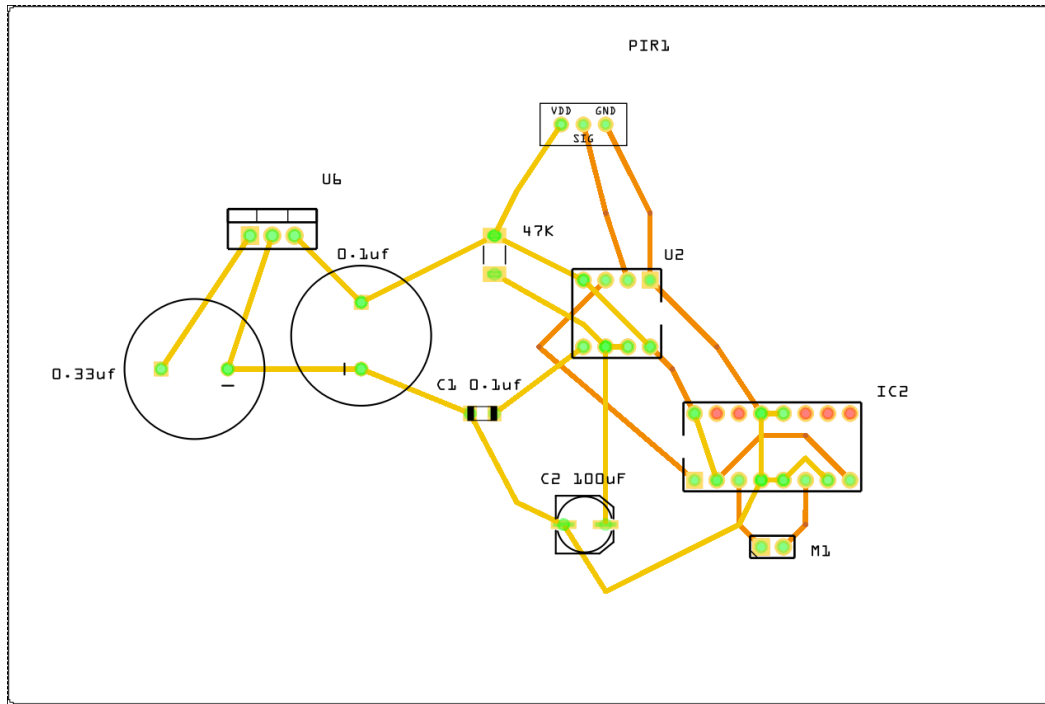
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Temperature-Controlled Fan PCB:



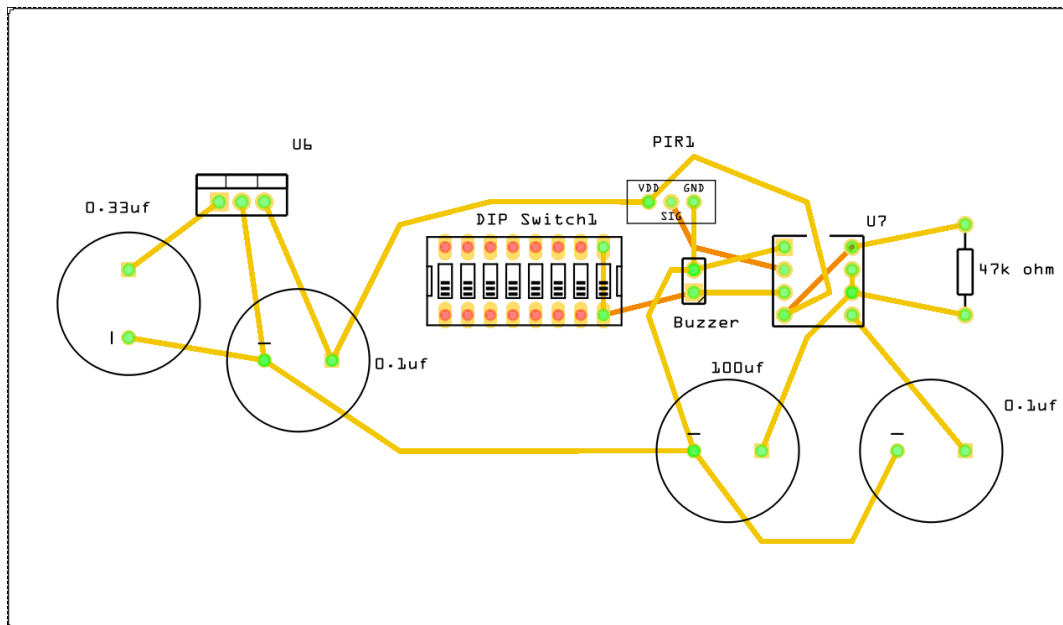
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Automatic Door PCB:



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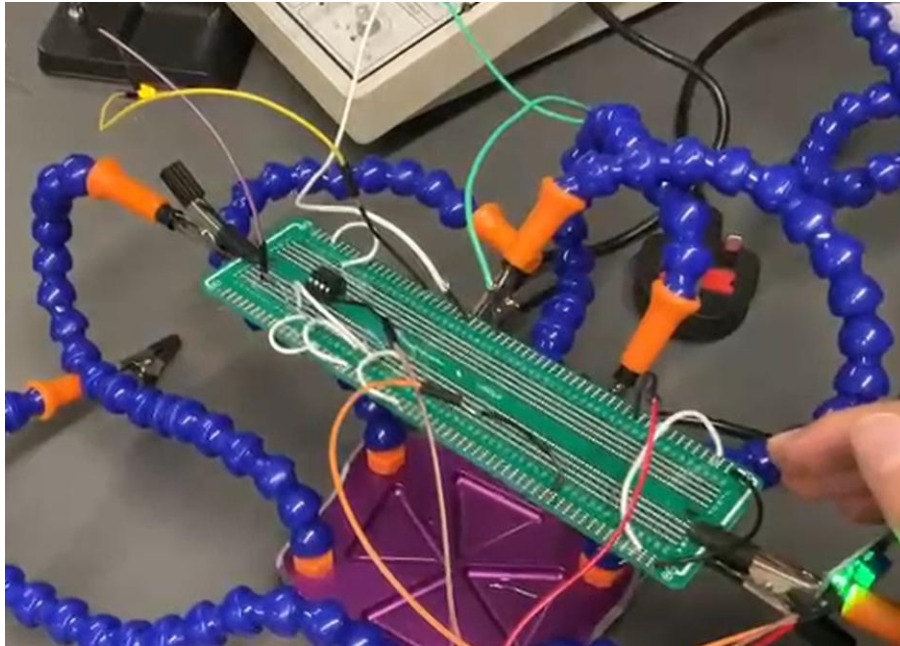
Security Alarm PCB:



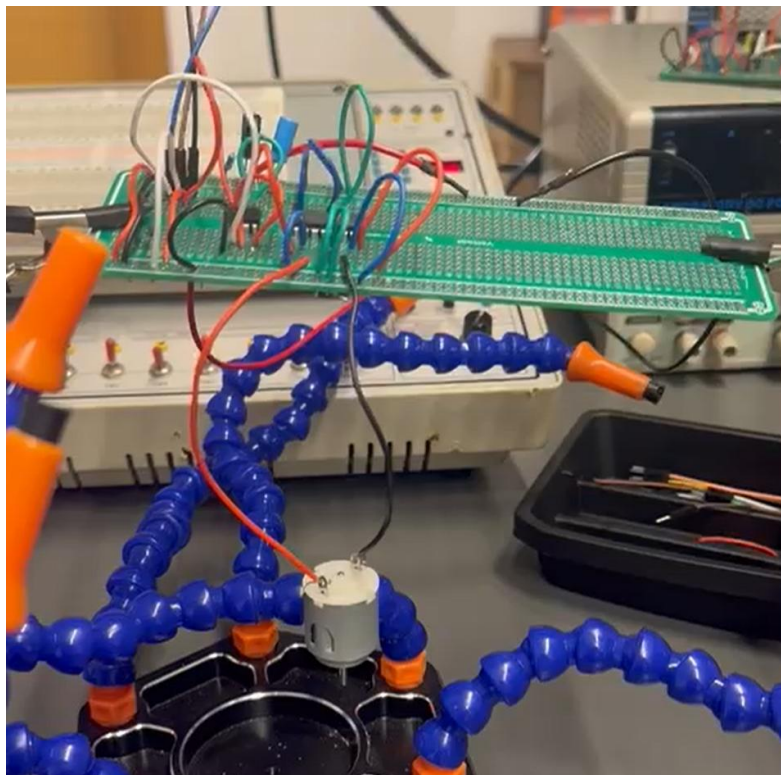
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Soldered Circuit & Prototype

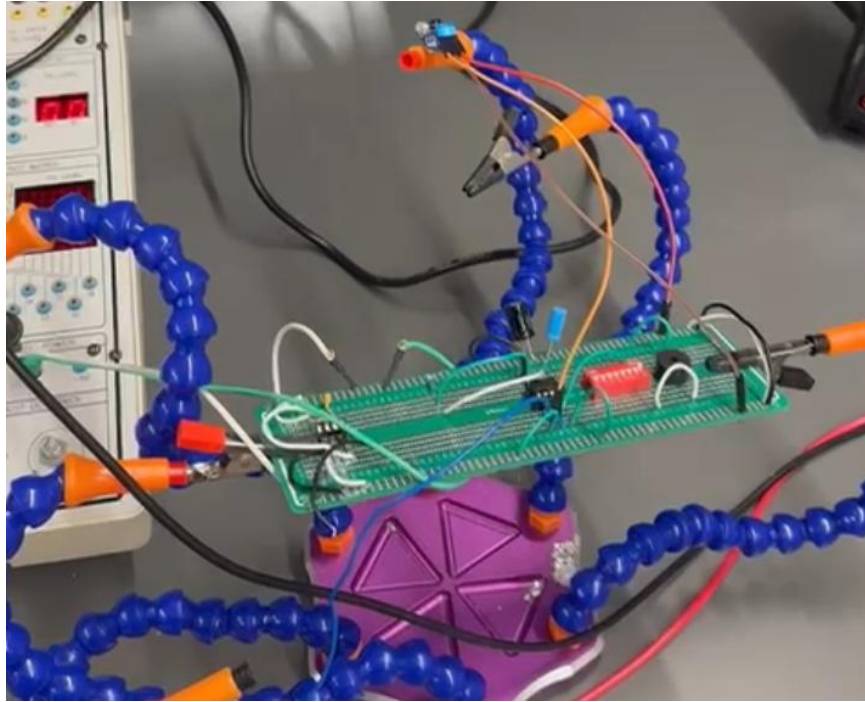
Automatic Lighting:



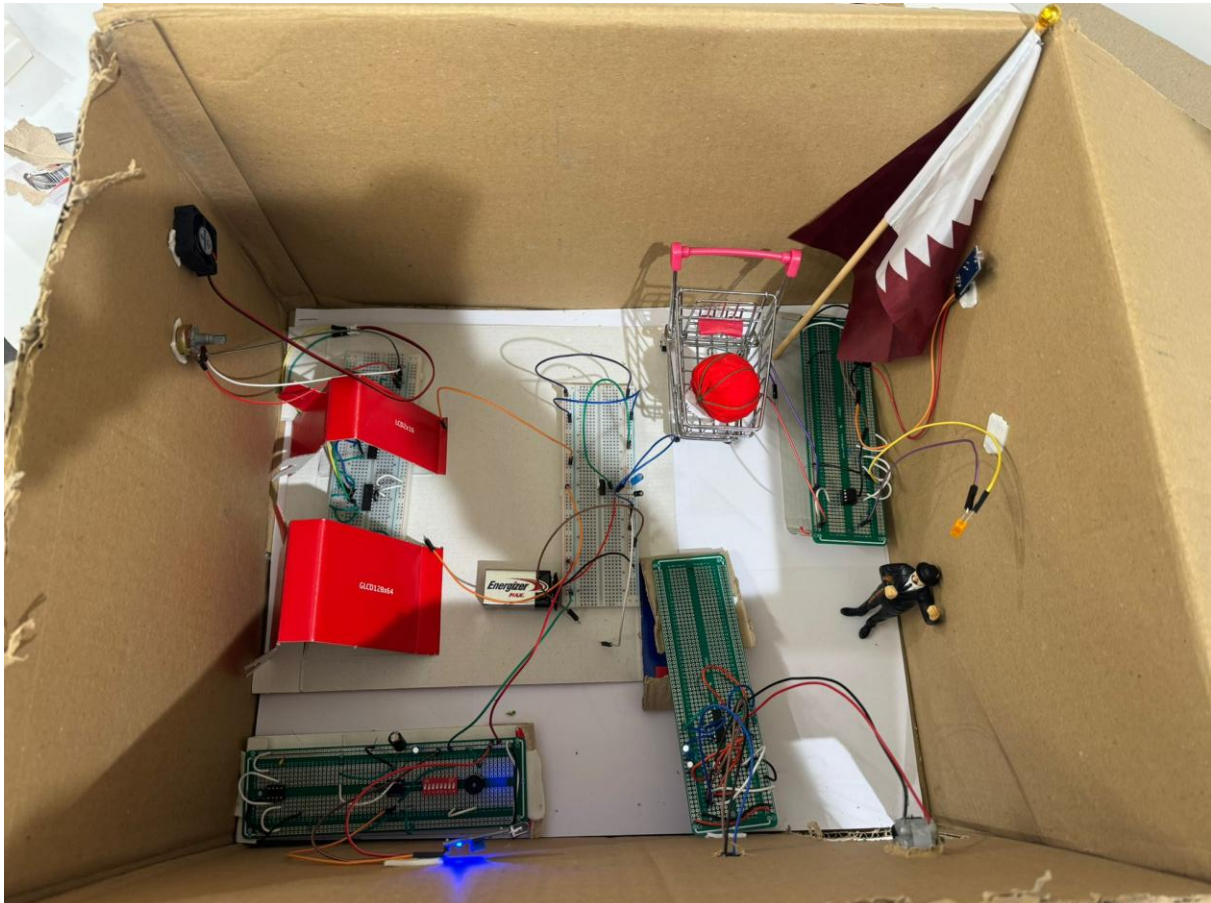
Automatic Door:



Security Alarm:



Final Prototype:



Discussion & Conclusion

Throughout this project, we underwent numerous difficulties. While implementing the circuits on a breadboard, we had trouble properly connecting them. We had to retrace every connection in order to finally connect them correctly. Another issue we ran into is that the alarm circuit had to be revised as its first version did not work as intended. However, the biggest trouble we ran into was during soldering. Even though the circuit was working while on a breadboard, after we soldered it would either not turn on, or would heat up and burn some components.

Although we faced some difficulties, we did not give up and instead worked hard to complete the project. When connecting circuits, we would pay extra attention to ensure each connection is properly connected. And when soldering we made sure not to repeat our mistakes, such as applying too much heat or creating solder bridges.

In conclusion, this project was a great learning experience for us. As we had to create all circuits ourselves, we now understand each component better and how we can use them to produce circuits that perform specific functions. We also learned how to create PCBs and bills of materials, preparing us for what we will need in the future. Every difficulty showed us where we had weaknesses, and how we can overcome them.